Data Recovery and Safety

Chapter 2: Common Storage Devices Common Problems & How To Fix Them

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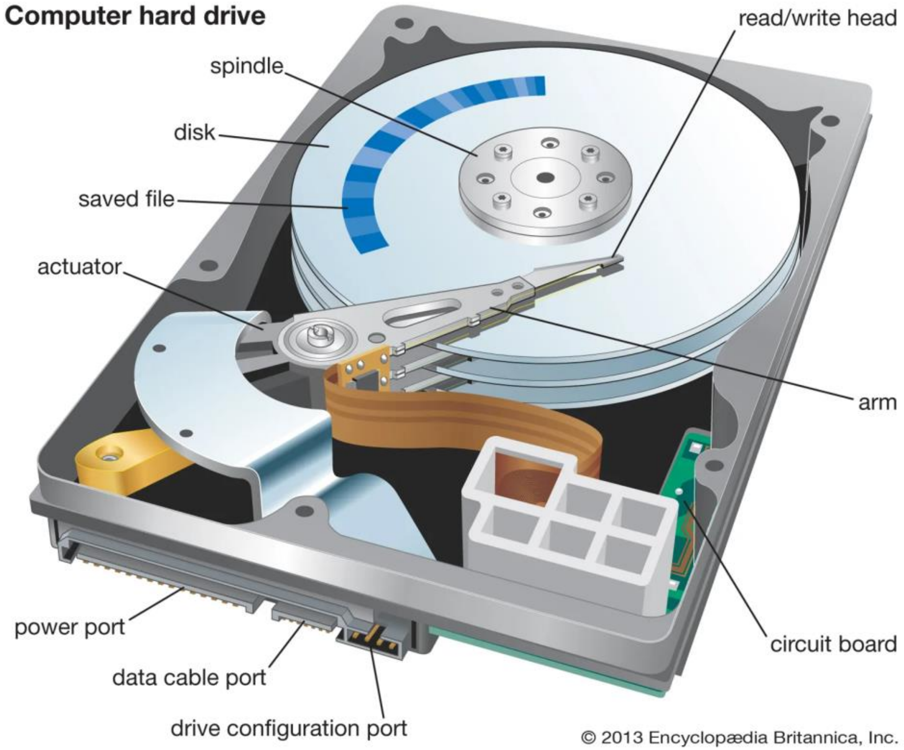
Common Storage Devices

Common Storage Devices

- **Internal HDD**: mainly found in computers, have the largest data storage capacity when compared to other devices. Now SSD and SSHD drives play a similar role.

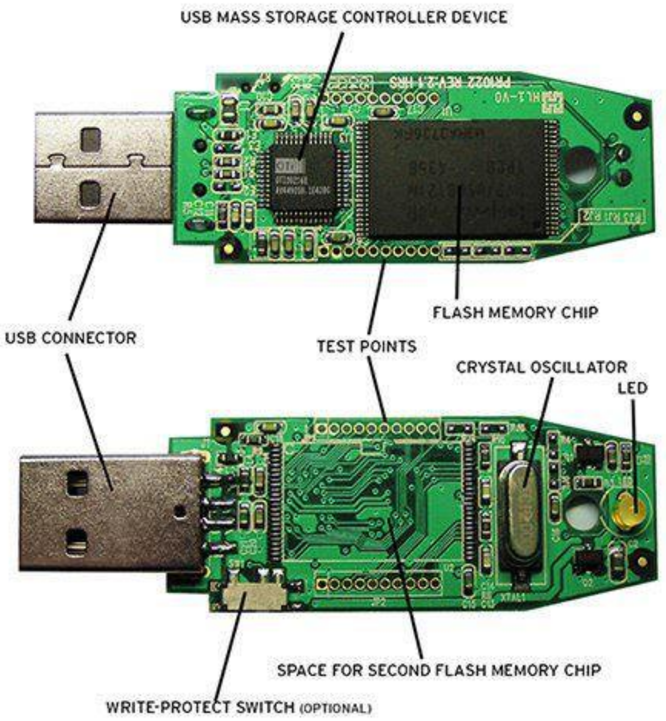


Common Storage Devices

- **External HDD:** External hard disk systems, their capacity is usually very large, so they can hold a lot of data, mainly used for backup or transfer.

Common Storage Devices

- **USB Flash:** Very popular due to its compact size, cheap & quite durable.
- **Memory Card:** very compact, mainly mou mainly mounted in small digital devi es (cell phones, cameras, …)



Common malfunction situations & Data recovery solutions

Common malfunction situations

Physical damage on Internal HDD:

- Service Area Failures: Quite a few cases of this error. In addition to objective reasons, users can cause SA errors in the situation of setting a password for HDD (stored on SA) and then forgetting it.
- Printed Circuit Board (PCB) Failure: Exposure to heat, dust and moisture, accidental impact (drops and falls), and power overloads/surges.
- The connection port or wire does not transmit the signal correctly (broken pins, physically damaged, incompatible,...)
- Power fluctuation
- Electro-mechanical and Electronics Component damaged/degraded
- Bad sector on disk

Common malfunction situations

Physical damage on External HDD:

- External HDDs are usually contained in HDD Boxes, the most common errors are:
  - Signal Conversion Board on the Box is faulty.
  - The communication port or connection wire is broken or incompatible.
  - The amperage is too low.
  - Other errors are the same as on Internal HDD.

Common malfunction situations

Physical damage on SSD:

- SSD Controller Failure
- Flash-memory chip failure
- Motherboard error
- The connection port or wire does not transmit the signal correctly (broken pins, physically damaged, incompatible,...)
- Power fluctuation
- Firmware error

Common malfunction situations

Physical damage on USB Flash Disk:

- Communication IC failure
- Flash-memory chip error
- Broken power supply fuse
- Broken pins, physically damaged
- Voltage stabilizer circuit failure
- Broken quartz
- Firmware error

Data recovery solutions

Physical damage on Internal HDD:

- Hard disk controller failure:
  - It is a very common error, causing loss of all data.
  - Usually, the telltale sign of this error is that the HDD cannot be detected.
  - Replacing a compatible board will save 100% of the data. Performing this operation is also very simple and does not affect the durability of the hard drive.
  - The key problem is how to find a compatible board. If you can’t find it, the board must be repaired, and this is very difficult.

Data recovery solutions

Physical damage on Internal HDD:

- Connection error:
  - It is possible that the connection wire (PATA/SATA1/SATA2/SATA3) is broken (inside) or oxidized, dusty, etc. The solution is to change the wire, because the wire is difficult to fix but very cheap.
  - Disk's connection port is faulty: Clean the dirt at the contact pins (with board cleaning solution).
  - Mother Board’s connection port is faulty: Prioritize trying another port (or another machine) and back up the data, then consider cleaning and fixing the broken port.

Data recovery solutions

Physical damage on Internal HDD:

- Power fluctuation:
    - PSU components or power circuits will degrade over time, causing voltage or amperage to be no longer suitable causing traceability errors.
    - The telltale signs of this problem are the disk performance is sluggish, the computer is slow, or even won't boot.
    - The first solution is to mount the HDD through another machine (maybe through the HDD Box) and backup data, because this error affects the durability of the HDD.

Data recovery solutions

Physical damage on Internal HDD

- Mechanical error:
  - Telltale signs: The drive makes strange sounds or feels no vibration when touched, showing that the reader has collided with the disc surface or the plate rotation motor has failed. If the sound is separate & short, the mechanical part of the reader assembly is damaged.
  - Solution: remove the HDD & replace the motor or reader assembly (or move the platters to another drive - easier), or put the platter in a dedicated reader.

Data recovery solutions

Physical damage on Internal HDD:

- Magnetic field error:
  - Telltale signs: some data or program collapsed. To confirm, just run the program to check the disk status for bad sectors or not. In the worst case, the operating system may refuse to access the disk.
  - Solution: For mild cases, the disk repair tools will repair themselves and “barrier” the bad sectors. In the worst cases, we have to access the sector at the direct level to get the data (no hardware intervention).

Data recovery solutions

Physical damage on Internal HDD

- Firmware error:
    - Solution: Reinstall the corresponding firmware (need to specify the exact model and manufacturer's name)

Data recovery solutions

Physical damage on External HDD

- Signal Conversion Board on the Box is faulty:
    - A very common error, the telltale sign is that the drive heats when connected to the computer, but the operating system does not recognize it or reports an error.
    - Solution: Open the Box, take out the internal HDD and move it to another Box or attach it inside the machine as Internal HDD (in most cases, it is okay). In special cases, the disk controller board must be replaced.

Data recovery solutions

Physical damage on External HDD:

- Connection port or wire error:
  - Errors often occur when connecting a USB 3.0 cable to a USB 2.0 port. The common reason is that the connection to the connection port is not secure, or the device does not recognize the device.
  - Solution: Replace with another connection cable (good one) and/or attach via USB 3.0 port. It is also possible to use a connection solution, such as Internal HDD.

Data recovery solutions

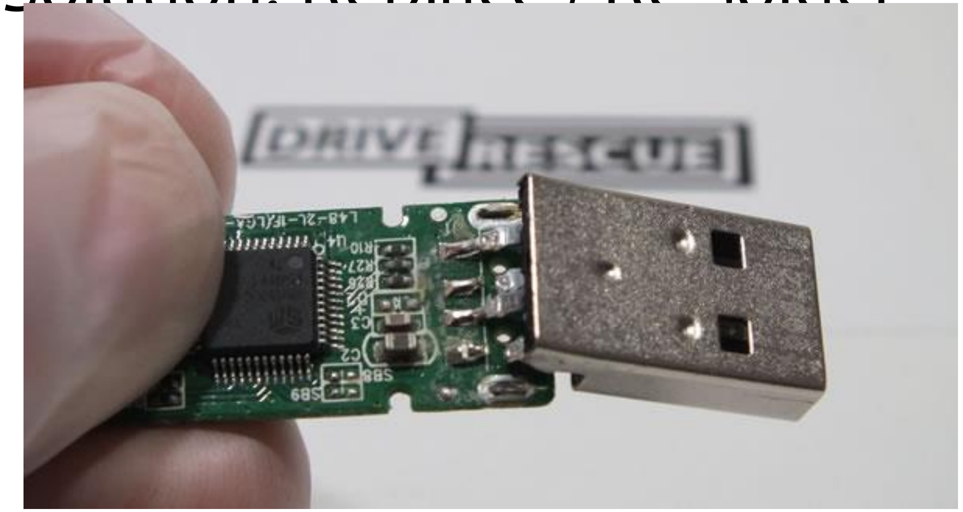
Physical damage on External HDD:

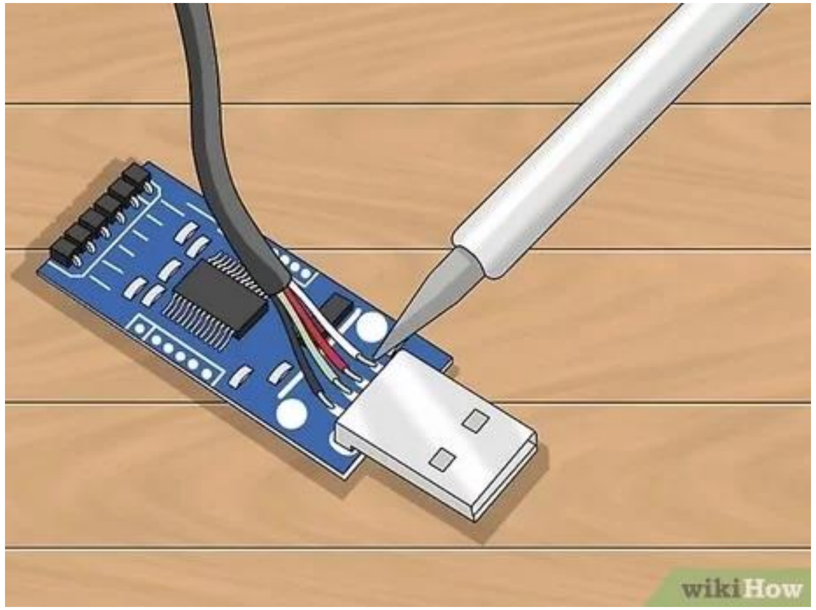
- The amperage is too low:
  - HDD Box is usually 2.5 inch in size and does not need auxiliary power (power from USB port / 1394). Quite a few machines have amperage at the ports not exceeding 500mA, while large capacity HDDs often need more than 500mA.
  - Solution: Besides connecting as an Internal HDD, it can be powered by a dedicated switch port or a 3-way wire (with an extra end to get more power from a USB port), or simply attach it to a port with enough power.

Data recovery solutions

Physical damage on USB Flash Disk:

- Damaged IC/contact pins, damaged power supply fuse, quartz:
- Solution: Replace / Re-solder





Data recovery solutions

Physical damage on USB Flash Disk:

- Firmware error:
  - Solution: Reinstall the corresponding firmware (need to specify the exact manufacturer name and controller chip model)
  - (You can use tools like ChipGenius, UsbFlashInfo, etc. to determine the Controller and Memory Chip codes - then use these two codes to search for the Firmware and the corresponding installation tool)

Experiments and exercises

Experiments

1. Accurately distinguish the above devices, explore the connection forms to retrieve data.
2. Cleaning contact pins, power supplies, direct connection port/wire.
3. Open and examine the reader, platter and accessories inside the hard drive and how to replace it.

Exercises

1. Make a table comparing the key characteristics of the above devices.
2. Make a detailed comparison table of HDD, SSD and SSHD
3. Experiment on opening/mounting the HDD board (give students a hard disk to take home to work)
4. Learn how to switch back and forth between the logical sector and the physical sector.

Projects:

1. Learn and show the organization of Partition on disk. Compare the advantages and disadvantages between MBR and GPT.
2. Write a program that directly accesses the logical/physical sector, displays the information obtained from the Parttion/BootSector table (if possible, implements the same interface as WinHex) and creates an Image for the disk to be repaired (the program needs to be able to resume unfinished tasks on its own in the event of an interruption – for example, a power failure.)

Questions and Answers